

Part 2 — Running the crawler on billions of URLs

Summary

Part 1 is a service that takes one URL and returns the page's metadata, its type and its topics. Part 2 is how to run that same code over a customer's monthly list of billions of URLs, store what comes back, and serve it to millions of requests, at a cost that can be defended.

The design is a pipeline. The list is loaded once a month into a **frontier**: a table of every waiting URL with a priority. A **scheduler** releases URLs from the frontier only as fast as each website allows, so the crawler is never rude to a host. A fleet of **workers** runs the Part 1 code on each released URL. Raw HTML goes to object storage; the extracted metadata and body go to a column-oriented database that serves both the customer's queries and single-URL lookups.

Two facts shape everything else:

- **Which URLs you have decides what limits you.** If the billions of URLs are spread across a million websites, the limit is CPU for parsing and the design is about fleet size. If they are concentrated on three websites, as the brief's own example says, the limit is how fast those three sites will let you fetch, and no fleet size changes that. Then the design is about choosing which fraction of the list to spend the budget on.
- **The parse is the cost.** Fetching a page waits on the network and costs almost nothing; reading it costs half a second of CPU. Every cost lever in this document is a way to parse fewer pages or parse them cheaper.

Rounded estimates, at 10 billion URLs a month across many sites: about 260 servers, about \$85,000 a month at the start rising to about \$120,000 by the second year as storage accumulates, roughly \$8 per million URLs. For the three-site example the same machinery runs on two servers and a few thousand dollars a month, and fetches about 78 million pages a month per site until the sites agree to more. Every figure is an estimate to within about $\pm 50\%$ until the proof of concept measures page size, parse cost and how often pages change. The exact arithmetic is in Appendix B and is generated by a script so it cannot drift from its inputs.

1. What the brief asks for

The brief asks for "a service that will take billions of URL and identify the metadata on the page, allow millions of requests on the content, and optimize for cost, performance and availability". The input is "a list of billions of URLs send in via a text file and/or in MySQL for a given year month", with the example "billions of URLs for amazon.com, walmart.com, bestbuy.com etc for July". The required outputs are a storage design for metadata and content, a unified schema, SLOs and SLAs, and the monitoring metrics and tools.

This document uses **host** for what the brief calls a domain, **list** for what the customer sends, **batch** for one month's run of that list, and **fetch** for retrieving one URL.

2. Two kinds of input

	Many hosts	Few hosts (the brief's example)
Shape of the list	billions of URLs over roughly a million websites	billions of URLs over three websites

	Many hosts	Few hosts (the brief's example)
What limits throughput	CPU to parse pages	the request rate each website tolerates
What the design must do	size and feed a fleet	spend a fixed per-site budget on the most valuable URLs, and report what was not reached

The arithmetic for the few-host case is short. A site that has not agreed to anything tolerates on the order of 10 requests a second from one crawler; that is an assumption, at the upper end of common practice. Ten a second is 26 million fetches a month. Against 3.3 billion URLs per site, that is under 1% of the list each month, and the whole list would take over ten years. At a negotiated 100 requests a second it is still over a year. No number of servers changes this. In that regime the system's job is to pick the right 1% and to tell the customer the coverage it achieved; §4.6 designs the picking.

The many-host case is the one that needs a fleet, and it is sized in §3.

3. How big is it

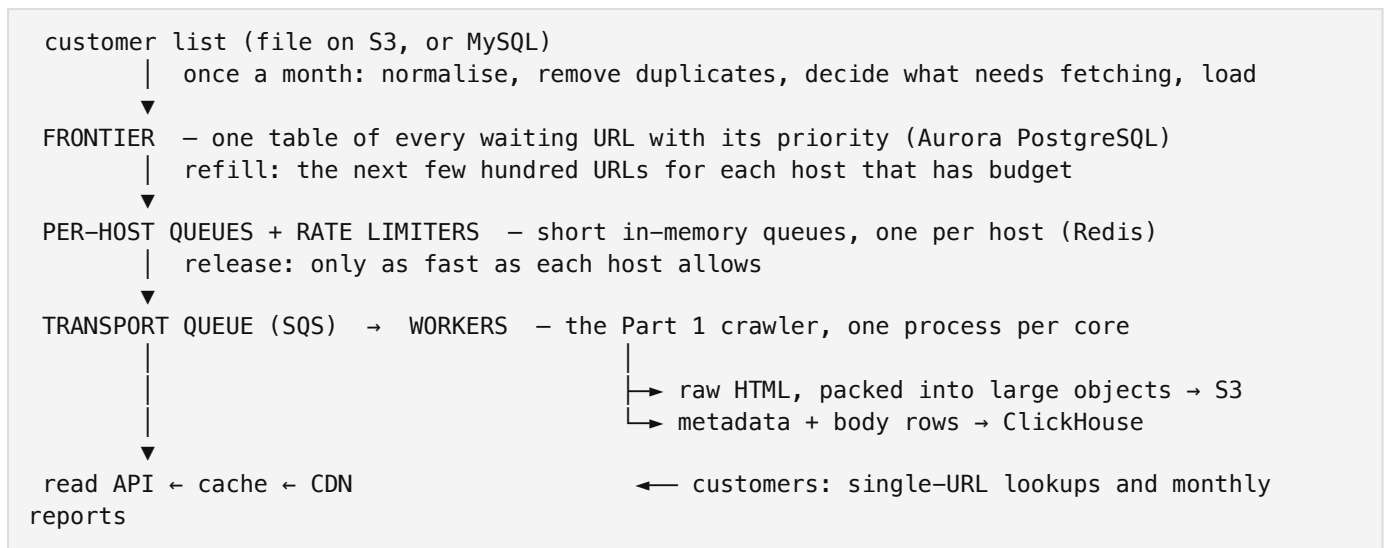
Planning volume: 10 billion URLs a month, an assumed upper case; at 2 billion, divide everything linear by five.

Quantity	Estimate	Basis
Fetches per second, sustained	~3,900	10 billion over 30 days
CPU per page	0.5 s measured on a laptop; 0.75 s assumed for a server core	measured on the two large fixture pages, five runs each
Cores needed	~2,900	fetch rate × CPU per page
Servers deployed	~260 of 16 cores, run at 70% so there is headroom	
Raw HTML stored per month	~400 TB compressed	200 KB a page assumed, compressing 5:1 as measured
Metadata and body stored per month	~25 TB compressed	~10 KB a record measured, compressing 4:1 assumed
Frontier table	~1.2 TB	10 billion rows of ~120 bytes

The least certain number is page size. The two fixture pages are 1.6 MB and 5.6 MB, far above the 200 KB planning value, and if the real average is 1 MB then storage and bandwidth are five times larger. The second least certain is parse cost on the real server. Both are the first things the proof of concept measures.

Memory is not a constraint: a worker process uses about 400 MB after a parse, so sixteen of them fit in a 32 GB server with room to spare.

4. The system



4.1 How a URL travels

1. The list arrives as a file or a database table. Each URL is normalised (lower-case host, tracking parameters removed) and hashed. Exact duplicates within the list are dropped by sorting. Each URL is compared with what has already been crawled to decide whether it needs fetching this month, and is given a priority. The result is loaded into the frontier table.
2. The scheduler looks at hosts that have budget and pulls their next few hundred URLs by priority into that host's short in-memory queue.
3. The scheduler releases URLs from those queues onto the transport queue, but only as many as each host's rate limiter permits. The transport queue therefore holds seconds of work, never a backlog for a host that is out of budget.
4. A worker takes a URL, checks the host's limiter once more, fetches the page, runs the Part 1 pipeline, appends the raw HTML to its current output object and writes the metadata row.
5. If a page turns out to be blocked, empty or an error page, that is recorded as a row too. Nothing is silently dropped.

4.2 The frontier

The frontier is where URLs wait. It is a table in Aurora PostgreSQL, partitioned by host, with one row per URL: host, URL, its hash, the customer's weight for it, when it is next due, and when it was last released. It is about 1.2 TB for 10 billion rows.

Why a table and not a queue: the order URLs should be fetched in changes over time. A page that changed last time should come back sooner; a page that has not changed in months can wait. A queue is fixed once written. A table lets the order be a query: "for this host, the URLs that are due, most valuable first". The due date is written once per fetch, so nothing is rewritten in bulk; priority is computed at query time from the stored weight and due date.

Why the in-memory per-host queues in front of it: a database cannot be queried 3,900 times a second for "next URL for this host". So the scheduler pulls a few hundred URLs per host at a time into Redis, and releases from there. Redis also holds each host's rate limiter, a token bucket refilled at the host's permitted rate. Both are rebuildable from the table, so losing Redis loses no URLs: anything released more than an hour ago without a result is released again, and fetching a URL twice is harmless.

The per-host queue is kept short, at most ten minutes of that host's rate, so a million hosts hold a few gigabytes in total, not a hundred.

This is the URL-frontier pattern from Alex Xu's *System Design Interview* (vol. 1, ch. 9): a prioritiser, priority-ordered front queues, one back queue per host for politeness, and a selector, with most of the frontier on disk and only the

heads in memory. The one deliberate difference is that the disk tier is a table rather than files, so priorities can change after loading.

4.3 Politeness

Every host has a rate cap: its `robots.txt` crawl delay, a rate the customer negotiated with the site, or the default of 10 requests a second. The limiter enforces it at release, and the worker checks it again immediately before fetching, so a backlog in the transport queue cannot be drained onto one host in a burst. If a host answers 429 or starts failing, its rate is halved and recovers slowly. `robots.txt` is fetched once per host, cached for a day, and obeyed per URL. Blocks are recorded, never evaded: the crawler identifies itself and does not pretend to be a browser.

4.4 Workers

A worker server runs one process per core, each with its own event loop and a few fetches in flight. Fetching is waiting on the network and needs few connections; parsing holds the CPU for half a second. Measured on the fixture pages, running the parse in a thread keeps the event loop responsive and gives some parallelism, but not enough to change the sizing; one process per core is the safe unit. Each server also runs a local DNS cache, because 3,900 lookups a second across a million hosts would otherwise saturate the cloud resolver.

Servers are Graviton Spot instances. Spot is about a third of the on-demand price, and a Spot interruption costs nothing here because every URL is an idempotent unit of work: fetch it again and you get the same row.

4.5 Storage

Three kinds of data, three stores, chosen by access pattern.

Data	Access pattern	Store
Raw HTML	written once, read rarely (re-extraction)	S3, packed 1,000 pages per object, tiered to cheaper classes with age
Metadata and extracted body	scanned by host and month; looked up by URL	ClickHouse, two tables (below)
Frontier, hosts, rate policies	transactional	Aurora PostgreSQL

Raw HTML is kept so the extractor can be re-run when it improves, which is far cheaper than fetching again. Each worker packs the pages it fetches into one object per minute; a page is addressed by object key, offset and length. Packing matters: one object per page would be 3,900 writes a second and about \$50,000 a month in request charges alone; packed, it is a few writes a second and about \$50.

Unified schema. One table holds every page type; product-specific fields such as price or SKU go in a key-value column rather than separate tables. The important columns:

Column	Meaning
<code>batch_month</code> , <code>fetches_at</code>	the month the customer asked for, and when the fetch actually happened
<code>host</code> , <code>url_hash</code> , <code>url</code> , <code>final_url</code>	identity, before and after redirects
<code>status</code> , <code>reason</code> , <code>robots_state</code> , <code>http_status</code>	what happened: success, blocked or error, and why
<code>title</code> , <code>description</code> , <code>canonical_url</code> , <code>author</code> , <code>published_date</code> , <code>language</code> , headings, Open Graph and Twitter tags	the metadata

Column	Meaning
body, word_count	the extracted content
page_type, topics (ranked)	classification and topics
s3_key, s3_offset, s3_length	where the raw HTML is
content_hash, extractor_version	for change detection and re-extraction
extra (map)	type-specific fields

Two ClickHouse tables carry it. `page_fetch` keeps every fetch, including blocked and failed ones, partitioned by day; it is the history and the audit trail. `page_current` keeps the latest row per URL, fed automatically from the first, partitioned by URL hash so repeat fetches collapse onto one row. Both are sorted by host then URL hash, which serves the customer's "all of amazon.com for July" scan and the "this one URL" lookup from the same store. Full definitions are in Appendix A.

4.6 Not fetching what you don't need

Three separate questions, answered exactly and without a Bloom filter (a Bloom filter at 1% false positives would silently drop a hundred million URLs a month as "already seen"):

- **Seen in this list?** Sort by hash at load time, drop neighbours.
- **Crawled before, and due again?** Join the list against `page_current` at load time.
- **Changed since last time?** After the fetch, hash the raw bytes; if equal to last time, skip the parse and the write. This is the largest saving in the design, because the parse is the cost. Its size depends on how often pages actually change: the planning assumption is 60% unchanged, which would remove about 60% of compute, but Amazon served a different variant of the same product page on every one of three fetches, so for hosts like that the saving may be near zero. The proof of concept measures it per host.

Choosing what to fetch when the budget is a fraction of the list (the few-host case): each URL carries the customer's weight and a due date. The due interval starts from the page type (articles weekly, products monthly), halves when the *extracted* content changed and doubles when it did not, within one day to ninety. Release order is by weight then due date. New URLs and recrawls share the budget half and half by default so a large new list does not starve freshness. Fixed schedules such as "news hourly" are not used: at any budget they starve the rest of the host.

4.7 Serving millions of requests

Assume 10 million reads a day, about 120 a second, twice that at peak. Two kinds: a single-URL lookup and an analytical report. Both go to ClickHouse; a single-URL lookup on the sorted table reads one small block. A Redis cache in front takes 90% of repeats, and a CDN caches and terminates TLS at the edge. A separate key-value copy of every record was considered and rejected: writing 10 billion records into DynamoDB would cost about \$125,000 a month to serve roughly a dozen cache misses a second. If the measured lookup latency is unacceptable, a key-value tier of one-kilobyte summaries returns at a tenth of that cost.

4.8 Pages that are not the page

Three measured facts from the test URLs: Amazon served the product page; CNN served the article; REI answered 403 for the page and for `robots.txt`, while serving the page to a browser's User-Agent. Coverage therefore has to be measured per host, and the design keeps two things apart everywhere: **blocked** (the site refused) is a coverage number; **error** (something failed) is a reliability number.

Some blocks are disguised. Myntra served a "Site Maintenance" page with a 200 to the crawler and the product page to a browser. The Part 1 code catches these generically: a 200 with almost no visible text and no metadata is

reported as `no_content`. At fleet scale two more signals separate a disguised block from a real outage: the same content hash on many URLs of one host is a template page, and a browser-User-Agent probe that succeeds where the crawler failed is a block.

Pages that need JavaScript to render are excluded in the first version and counted: a sampled detector reports the share per host, and rendering, at roughly ten to twenty times the CPU of a static parse, is budgeted only if that share is material. SLAs are written in terms of fetchable pages and report the excluded counts.

5. When things fail

Everything is idempotent, so the answer to almost every failure is "do it again".

Failure	What happens
Worker dies mid-fetch	the queue message reappears; the URL is fetched again
Spot instance reclaimed	two minutes' notice: stop taking work, flush the output object, exit
Redis lost	limiters rebuilt from policy, queues refilled from the frontier; anything released over an hour ago without a result is released again
Frontier table unavailable	no refills; queues drain within minutes and the fleet idles visibly; nothing lost
A page crashes the parser	after three attempts it is dead-lettered and recorded as <code>parse_failed</code>
ClickHouse falls behind	workers buffer for a few minutes, then the scheduler slows releases
A host starts blocking	its rate halves; an alert and a coverage note to the customer
Bad deploy	roll back; deploys freeze while the error budget recovers

Back-pressure always flows toward the frontier, the one place that can slow the system without losing anything.

6. Objectives and commitments

Internal objectives (SLOs), measured continuously:

Objective	Many hosts	Few hosts
Released URL fetched	99% within 1 hour	same
Batch completion	99% of fetchable URLs within the month plus a day	at least 95% of each host's agreed daily budget used
Parse latency	95th percentile under 1.5 s	same
Permanent errors (not blocks)	under 0.5%	same
Row visible to queries	within 10 minutes of fetch	same
Read API	99.9% available; cached reads under 200 ms	same

External commitments (SLAs), deliberately looser: API 99.5% monthly; 95% of *fetchable* URLs per batch, with blocked, disallowed and JavaScript-only counts reported per host; for the few-host case, coverage at the agreed rate with a per-host estimate of completion rather than a percentage of the list. "Fetchable" is the operative word: promising a share of *all* URLs would be promising to defeat bot detection.

7. Monitoring

Per-host figures are computed in ClickHouse from the fetch table, not emitted as metric labels, because a million hosts would overwhelm a metrics database. Fleet metrics go to Prometheus and Grafana; queue and autoscaling signals to CloudWatch; a sample of fetches is traced end to end.

Metric	Threshold	Action
Batch progress and per-host completion estimate	estimate past the deadline	if fleet-limited, scale; if host-limited, report to the customer
Age of oldest released message	over 1 hour	add 20% capacity, up to 1.5× the mean
Block rate per host	2× its baseline	coverage ticket; customer report
Error rate per host	over 5% for 15 minutes	check host health; pause if 5xx
no_content rate per host; many URLs sharing one hash	over 1%; over 100 URLs	probe with a browser User-Agent; if it succeeds, count the host as blocking
Parse latency and parse failures	p95 over 1.5 s; failures over 1%	page size growth or a parser regression
Unchanged rate on recrawl	drops 10 points week on week	normalisation or hashing regression; cost rises directly
ClickHouse parts and insert lag	over 300 parts; lag over 5 minutes	larger insert batches; slow the scheduler
Cost per million URLs	over 20% above model	check the unchanged rate, page size, instance mix

Every worker writes one structured log event per fetch with the URL, reason, robots state and HTTP status; that event and the row it accompanies are the source for all of the above.

8. What it costs

Prices are public list prices for us-east-1 as recalled at writing; every figure below is an estimate to within about $\pm 50\%$ until the proof of concept replaces the assumptions with measurements. Exact arithmetic is in Appendix B.

Assumptions: 10 billion URLs a month; 200 KB per page, compressing 5:1; 0.75 s of CPU per page on a server core; Graviton Spot at \$0.02 per core-hour; 10 KB per metadata record; storage tiered with age.

Many hosts, rounded:

Line	Per month	Notes
Worker fleet, ~260 Spot servers	~\$60,000	the dominant line; scales with pages parsed
Databases, caches, ingest, API, monitoring	~\$10,000	fixed; ClickHouse with two replicas, Aurora with its I/O
Egress, queue, object writes	~\$2,000	after the two design choices below
Raw HTML storage	~\$9,000 in month 1, ~\$28,000 by month 24	a stock that accumulates; tiered to cold storage
Metadata storage	~\$4,000 in month 1, ~\$24,000 by month 24	hot tier replicated, cold tier on S3
Total	~\$85,000 in month 1; ~\$120,000 by month 24	about \$8 per million URLs at the start

Few hosts: the same components on two worker servers, a fixed floor of about \$2,000 a month, roughly \$27 per million URLs at 78 million a month. Cost is not the constraint in this regime; the sites' tolerance is.

The levers, which matter more than the totals:

Decision	Effect	Why
Skip the parse when the raw page is unchanged	up to -60% of the fleet	the parse is the cost; the saving is bounded by how often pages actually change
Pack raw pages into large objects	-99.9% of object-write charges	1,000 pages per write instead of one
Public IP egress instead of a NAT gateway	-95% of egress cost	NAT charges per byte; ~400 TB a month of compressed pages
Tier storage with age	-80% on data older than three months	raw HTML is read rarely after the re-extraction window
Bill instances at realistic utilisation	+40% versus a naive core count	servers are paid for whether busy or not; the model runs them at 70%
No key-value replica for serving	avoids ~\$125,000 a month	the sorted column store already answers single-URL lookups

9. What to validate first

In the order that would change the design most if wrong:

1. The request rate the target hosts accept, probed at 1, 5 and 10 a second. It decides the regime and everything promised.
2. Page-size distribution. Storage and bandwidth scale with it directly.
3. Parse cost on the real server type. It sets the fleet.
4. How often pages are unchanged on recrawl, per host. It sets the largest saving.
5. Block rate and JavaScript-only share per host. They set the coverage ceiling.
6. ClickHouse behaviour at 3,900 inserts a second, and single-URL lookup latency under that load.
7. Frontier refill latency at a million hosts on a 1.2 TB table, and the re-release path after a Redis failover.
8. A fair column-store versus row-store comparison on real hardware before anyone quotes a speed multiple.

Appendix A — Decisions and schema

Each decision names what it serves, what was chosen, the alternatives and the property that ruled each out, the condition that would reverse it, and what to measure first. None of these components has been operated by the author in production; the choices are architectural and each names its validation.

#	Decision	Chosen	Rejected, and why	Reverses if	Validate
1	Transport queue	SQS, batched	Kafka: ordering and replay unused, partitions would track hosts. RabbitMQ: a queue per host and paid empty polls	several consumers need the same event	receive latency at 3,900/s
2	Rate limiters and per-host queues	Redis, Lua token bucket	DynamoDB: a write per token, ~\$12,500/month and 10 ms. Postgres: a hot-row write per fetch	over ~50M hosts (shard)	tokens per node; failover

#	Decision	Chosen	Rejected, and why	Reverses if	Validate
3	Frontier store	Aurora PostgreSQL table, partitioned by host, priority computed at query time from stored weight and due date	sorted files on S3 read in order: the order cannot change after loading. All in Redis: ~1 TB of RAM	refill queries cannot keep up (pre-materialise heads)	refill latency at 1M hosts
4	Worker compute	Graviton Spot, one process per core	Lambda: ~\$64,000/month and no connection reuse. Fargate Spot: no control of process placement. EKS: a cluster to run an autoscaling group	Spot interruptions break the 1-hour objective (add on-demand baseline)	parse CPU on the target core
5	Raw store	S3, tiered	EFS: ~13× the price. HDFS: storage tied to compute	frequent re-extraction (keep more hot)	compression ratio across hosts
6	Raw object packing	per (worker, minute), 1,000 pages	per page: 3,900 writes/s and ~\$50,000/month. Per (host, hour): at a million hosts each buffer holds one page	pages are rewritten often (they are not)	ranged-read latency
7	Metadata store	ClickHouse, two tables, two replicas	Postgres: reads every column of every row. BigQuery: per-TB-scanned pricing across clouds. Elasticsearch: 2–3× storage, not a scanner. DynamoDB: no scans	insert rate produces unmanageable parts	inserts at 3,900 rows/s; lookup p99
8	Schema	one table for all types; extra map; history by day, current by URL hash	one table by month: 10 billion rows in one partition and recrawls never collapse. Partition by host: too many parts	history never queried (keep 90 days)	parts per partition
9	Dedup	exact sort and join at load	Bloom filter: 1% false positives drop 100M URLs/month and cannot forget. Redis set: ~500 GB	URLs arrive as a stream	join time for 10 billion rows
10	Serving	CDN → cache → ClickHouse	DynamoDB replica: ~\$125,000/month in writes	lookup p99 over 100 ms under load	that p99
11	Egress	public IPv4 on workers	NAT gateway: ~\$18,000/month per byte. NAT instances: single-AZ	a partner needs a short allow-list (NAT with fixed IPs)	block rate by egress IP
12	Observability	Prometheus + Grafana, CloudWatch, traces; per-host figures in ClickHouse	Datadog: per-host pricing. CloudWatch custom metrics per host: per-metric pricing × 1M	few-host regime (CloudWatch alone)	series count
13	JavaScript rendering	excluded and measured	render all: fleet × 10–20. Ignore: hidden coverage gap	measured share over 10%	that share
14	Recrawl policy	adaptive interval per URL inside a per-host budget	fixed cadence per type: "hourly" starves the rest of the host. Uniform monthly: no freshness	customer supplies priorities	change rate per type

Schema, ClickHouse:

```

CREATE TABLE page_fetch (
  batch_month Date, fetched_at DateTime64(3),
  host LowCardinality(String), url_hash UInt64, url String, final_url String,
  status LowCardinality(String), reason LowCardinality(String), robots_state
LowCardinality(String),
  http_status Nullable(UInt16), content_type LowCardinality(String), content_hash UInt64,
  title String, description String, canonical_url String, author String,
  published_date Nullable(DateTime), language LowCardinality(String), h1_headings
Array(String),
  og_tags Map(String, String), twitter_tags Map(String, String),
  body String CODEC(ZSTD(3)), word_count UInt32,
  page_type LowCardinality(String), topics Nested(topic String, rank_score Float32),
  s3_key String, s3_offset UInt32, s3_length UInt32,
  extractor_version LowCardinality(String), extra Map(String, String)
) ENGINE = MergeTree PARTITION BY toYYYYMMDD(fetched_at) ORDER BY (host, url_hash,
fetched_at);

CREATE TABLE page_current AS page_fetch
ENGINE = ReplacingMergeTree(fetched_at) PARTITION BY cityHash64(url_hash) % 64 ORDER BY (host,
url_hash);

```

Frontier, Aurora PostgreSQL: `frontier(host, url_hash, url, customer_weight, due_at, released_at, batch_month)`, partitioned by host hash, indexed on `(host, released_at, customer_weight DESC, due_at)`. Rows older than the batch after the current one are purged.

Appendix B — The numbers

The full number ledger (every measured, assumed and derived value with its working) is Appendix B of `docs/part2-scale-design.md` in the repository, generated and checked by `scripts/ledger.py`. It is omitted from the PDF for length.